MATH 160 — Logic & Propositions

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Name: _____ Date: _____ Score: / 10

Learning Objectives

- Build and interpret truth tables for ∧, ∨, ¬, →, and ↔
- Apply De Morgan's Laws and logical equivalences
- Evaluate compound propositions and identify tautologies
- Construct and validate logical arguments (Modus Ponens, contrapositive)

For truth tables: always list all 2^n rows for n variables. Use T and F consistently — not 1/0 or True/False.

1. Complete the truth table for p ∧ q (AND) and p ∨ q (OR).

p	q	p ∧ q	p ∨ q
T	T	____	____
T	F	____	____
F	T	____	____
F	F	____	____

Answer: _____

2. Complete the truth table for ¬p, p → q (IF p THEN q), and ¬p ∨ q.

p	q	¬p	p → q	¬p ∨ q
T	T	____	____	____
T	F	____	____	____
F	T	____	____	____
F	F	____	____	____

Answer: _____

3. Complete the truth table to verify De Morgan's Law: ¬(p ∧ q) ≡ ¬p ∨ ¬q.

p	q	p ∧ q	¬(p ∧ q)	¬p ∨ ¬q
T	T	____	____	____
T	F	____	____	____
F	T	____	____	____
F	F	____	____	____

Answer: _____

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4. Complete the truth table for $p \wedge (q \vee r)$.

p	q	r	q ∨ r	p ∧ (q ∨ r)
T	T	T	___	___
T	T	F	___	___
T	F	T	___	___
T	F	F	___	___
F	T	T	___	___
F	T	F	___	___
F	F	T	___	___
F	F	F	___	___

Answer: _____

5. Build a truth table for $(p \rightarrow q) \wedge (q \rightarrow p)$. Is this a tautology, contradiction, or neither?

p	q	p → q	q → p	(p→q) ∧ (q→p)
T	T	___	___	___
T	F	___	___	___
F	T	___	___	___
F	F	___	___	___

Answer: _____

6. Evaluate the logical argument: "If it rains, the ground is wet. It is raining. Therefore, the ground is wet." Is this valid?

$$p \rightarrow q, \quad p \vdash q$$

Answer: _____

7. Write the converse, inverse, and contrapositive of: "If it snows, then school is cancelled."

$$p \rightarrow q$$

Answer: _____

8. Determine the truth value of $(p \vee \neg q) \rightarrow (\neg p \wedge q)$ when p = T and q = F.

$$p = T,\ q = F \Rightarrow (p \vee \neg q) \rightarrow (\neg p \wedge q)$$

Answer: _____

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9. Complete the truth table for p ↔ q (biconditional, "p if and only if q").

p	q	p ↔ q	p → q	q → p
T	T	____	____	____
T	F	____	____	____
F	T	____	____	____
F	F	____	____	____

Answer: _____

10. Use a truth table to show whether p → q and ¬q → ¬p are logically equivalent.

p	q	¬p	¬q	p → q	¬q → ¬p
T	T	____	____	____	____
T	F	____	____	____	____
F	T	____	____	____	____
F	F	____	____	____	____

Answer: _____

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MATH 160 — Logic & Propositions

Problems 1–5 are full truth tables — key exam skill. Problem 3 verifies De Morgan's Law by table. Problem 7 introduces the contrapositive (only equivalent transform).

Solutions

1. Complete the truth table for p ∧ q (AND) and p ∨ q (OR).

p	q	p ∧ q	p ∨ q
T	T	___	___
T	F	___	___
F	T	___	___
F	F	___	___

→ *p ∧ q (AND) is T only when BOTH p and q are T.*
→ *p ∨ q (OR) is T when AT LEAST ONE of p, q is T.*
→ *Row 1: T∧T=T, T∨T=T. Row 2: T∧F=F, T∨F=T.*
→ *Row 3: F∧T=F, F∨T=T. Row 4: F∧F=F, F∨F=F.*

Answer:

p	q	p ∧ q	p ∨ q
T	T	T	**T**
T	F	F	**T**
F	T	F	**T**
F	F	F	**F**

2. Complete the truth table for ¬p, p → q (IF p THEN q), and ¬p ∨ q.

p	q	¬p	p → q	¬p ∨ q
T	T	___	___	___
T	F	___	___	___
F	T	___	___	___
F	F	___	___	___

→ *¬p flips the truth value of p.*
→ *p → q (conditional) is F only when p is T and q is F.*
→ *Key insight: p → q is logically equivalent to ¬p ∨ q (same column!).*
→ *This equivalence is used to prove theorems by contrapositive.*

Answer:

p	q	¬p	p → q	¬p ∨ q
T	T	F	T	**T**
T	F	F	F	**F**
F	T	T	T	**T**
F	F	T	T	**T**

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3. Complete the truth table to verify De Morgan's Law: $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

p	q	p ∧ q	¬(p ∧ q)	¬p ∨ ¬q
T	T	___	___	___
T	F	___	___	___
F	T	___	___	___
F	F	___	___	___

→ *Columns 4 and 5 are identical → the two expressions are logically equivalent.*
→ *De Morgan's Law 1: $\neg(p \wedge q) \equiv \neg p \vee \neg q$.*
→ *De Morgan's Law 2: $\neg(p \vee q) \equiv \neg p \wedge \neg q$.*
→ *These laws let you distribute negation across AND/OR.*

Answer:

p	q	p ∧ q	¬(p ∧ q)	¬p ∨ ¬q
T	T	T	F	F
T	F	F	T	T
F	T	F	T	T
F	F	F	T	T

4. Complete the truth table for $p \wedge (q \vee r)$.

p	q	r	q ∨ r	p ∧ (q ∨ r)
T	T	T	___	___
T	T	F	___	___
T	F	T	___	___
T	F	F	___	___
F	T	T	___	___
F	T	F	___	___
F	F	T	___	___
F	F	F	___	___

→ *First evaluate $q \vee r$ (OR) for each row.*
→ *Then evaluate $p \wedge (q \vee r)$ — AND with result from column 4.*
→ *$p \wedge (q \vee r)$ is T only when p is T AND at least one of q, r is T.*
→ *Rows 1–3 are T; row 4 (T,F,F) gives F because $q \vee r$ = F.*

Answer:

p	q	r	q ∨ r	p ∧ (q ∨ r)
T	T	T	T	T
T	T	F	T	T
T	F	T	T	T
T	F	F	F	F
F	T	T	T	F
F	T	F	T	F
F	F	T	T	F
F	F	F	F	F

5. Build a truth table for $(p \to q) \wedge (q \to p)$. Is this a tautology, contradiction, or neither?

p	q	p → q	q → p	(p→q) ∧ (q→p)
T	T	___	___	___
T	F	___	___	___
F	T	___	___	___
F	F	___	___	___

→ *$(p \to q) \wedge (q \to p)$ = T only in rows 1 and 4 (when p and q have the same value).*
→ *This is the biconditional $p \leftrightarrow q$ — neither always T (tautology) nor always F.*
→ *It is a contingency (sometimes T, sometimes F).*

Answer:

p	q	p → q	q → p	(p→q) ∧ (q→p)
T	T	T	T	**T**
T	F	F	T	**F**
F	T	T	F	**F**
F	F	T	T	**T**

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6. Evaluate the logical argument: "If it rains, the ground is wet. It is raining. Therefore, the ground is wet." Is this valid?

$$p \to q, \quad p \vdash q$$

→ *Let p = "It is raining", q = "The ground is wet".*
→ *Premise 1: $p \to q$. Premise 2: p.*
→ *Modus Ponens rule: from $p \to q$ and p, we can conclude q.*
→ *The argument form is valid (always produces T when premises are T).*

Answer: Valid — Modus Ponens

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7. Write the converse, inverse, and contrapositive of: "If it snows, then school is cancelled."

$$p \to q$$

→ *Original ($p \to q$): "If it snows, school is cancelled."*
→ *Converse ($q \to p$): "If school is cancelled, then it snowed." (NOT equivalent)*
→ *Inverse ($\neg p \to \neg q$): "If it does not snow, school is not cancelled." (NOT equivalent)*
→ *Contrapositive ($\neg q \to \neg p$): "If school is not cancelled, it did not snow." (EQUIVALENT)*
→ *Only the contrapositive is logically equivalent to the original.*

Answer: Contrapositive: $\neg q \to \neg p \equiv p \to q$

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8. Determine the truth value of $(p \vee \neg q) \to (\neg p \wedge q)$ when p = T and q = F.

$$p = T,\ q = F \Rightarrow (p \vee \neg q) \to (\neg p \wedge q)$$

→ *$\neg q = \neg F = T$. $p \vee \neg q = T \vee T = T$.*
→ *$\neg p = \neg T = F$. $\neg p \wedge q = F \wedge F = F$.*
→ *$(p \vee \neg q) \to (\neg p \wedge q) = T \to F = F$.*
→ *A conditional is F only when the hypothesis is T and conclusion is F.*

Answer: $T \to F = F$

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9. Complete the truth table for $p \leftrightarrow q$ (biconditional, "p if and only if q").

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$
T	T	___	___	___
T	F	___	___	___
F	T	___	___	___
F	F	___	___	___

→ *$p \leftrightarrow q$ is T when p and q have the SAME truth value.*
→ *Equivalently: $p \leftrightarrow q = (p \rightarrow q) \wedge (q \rightarrow p)$.*
→ *Notice: $p \leftrightarrow q$ column matches rows where p=q.*

Answer:

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$
T	T	T	T	**T**
T	F	F	F	**T**
F	T	F	T	**F**
F	F	T	T	**T**

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10. Use a truth table to show whether $p \rightarrow q$ and $\neg q \rightarrow \neg p$ are logically equivalent.

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	___	___	___	___
T	F	___	___	___	___
F	T	___	___	___	___
F	F	___	___	___	___

→ *Columns 5 and 6 are identical → $p \rightarrow q \equiv \neg q \rightarrow \neg p$.*
→ *$\neg q \rightarrow \neg p$ is the CONTRAPOSITIVE of $p \rightarrow q$.*
→ *A conditional and its contrapositive are always logically equivalent.*

Answer:

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	**T**
T	F	F	T	F	**F**
F	T	T	F	T	**T**
F	F	T	T	T	**T**